A Conditional Reduction of Twin-Prime Abundance to a Signed Ford-Weighted Exterior Estimate

The direct-scalar endgame and the degree-twelve auxiliary reduction

Research note prepared from the supplied Gate 1B / Real R11 records
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Abstract

We give a source-explicit conditional argument leading to a lower bound of order $X/(\log X)^2$ for the number of primes $p \in (X/2, X]$ for which $p+2$ is prime. The arithmetic input is a one-sided estimate for a specified signed exterior sum built from the Ford–Maynard sieve weight and a fixed rough comparison sequence. We prove the finite partition identity, the sign-sensitive endgame, and the removal of proper prime powers. We also prove that the project's proposed rough-input transfer and twelfth-moment estimate would reduce the outstanding exterior estimate to a single, explicitly defined sum $\mathcal{J}_{12}(X)$. The latter estimate is not proved here.

The manuscript separates elementary deductions proved below, published mathematical inputs, project-specific analytic deductions awaiting independent review, and the genuinely open correlation estimate. In particular, it does not assert that a restricted R11 bound, an ownership table, or a conditional Lean declaration proves the Twin Prime Conjecture. No claim of historical novelty, completed formal verification, or prize eligibility is made.

Central conditional implication. For the *same* source throughout, the supporting estimates in Hypothesis 3.3, together with

$$S_X \le \left(\frac{c_F}{4} - \rho\right) B_*(X) \qquad (\rho \ge 0),$$

imply

$$\#\{p \in (X/2, X] : p, p+2 \in \mathbb{P}\} \ge \frac{c_F b_0}{8} \frac{X}{(\log X)^2}$$

for all sufficiently large X. The constants c_F, b_0 are positive and fixed. The degree-twelve route replaces the missing bound on S_X by

$$\boxed{\limsup_{X\to\infty} \frac{\mathcal{J}_{12}(X)}{B_*(X)} \le \frac{c_F}{8}}$$

only after its two explicitly stated transfer hypotheses have been verified. This boxed assertion remains open in the supplied records.

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1 Scope, evidence, and reading conventions

This note concerns the direct exterior-sum route in the supplied Gate 1B / Real R11 project. It is not a survey of all branches of that project. The purpose is to put the conditional implication into ordinary mathematical notation so that a reader need not understand internal labels or software declarations.

A statement marked *proved here* is established by the displayed argument. A *project hypothesis* is a precise mathematical assertion needed for an application; a proof sketch or reported computer check in a source record does not, in this note, promote it to an independently verified theorem. Public results are attributed to their authors. These categories must not be merged.

The main endgame is proved in Section 3. The additional requirements of the current degree-twelve route are in Sections 5–6. Section 8 lists the remaining obligations. An alternative conditional R11/Gram export is included in Section 7; it is *not* a prerequisite for the direct route.

All logarithms are natural. All Euler products are over primes. Every occurrence of "sufficiently large" permits a threshold depending on previously fixed parameters. In particular, $\varepsilon > 0$ is fixed before $X \to \infty$. No effective numerical threshold is supplied. If an earlier source uses a different weight, endpoint convention, comparison, or normalization, its estimate cannot be substituted without a proved map to the definitions below.

2 The arithmetic source, defined explicitly

2.1 Prime detector and comparison

For $X > e^2$, put

$$L = \log X, \qquad I_X = (X/2, X] \cap \mathbb{N}, \qquad \mathcal{I}_X = \{n \in I_X : n \text{ is odd}\}. \tag{2.1}$$

The functions μ, τ, ω and φ denote, respectively, the Möbius function, the divisor-counting function, the number of distinct prime factors, and Euler's totient. We use $P^-(1) = +\infty$. The symbol Λ denotes the von Mangoldt function: $\Lambda(q^a) = \log q$ for a prime q and an integer $a \geq 1$, and it is zero otherwise. Let

$$z = \lfloor e^{\sqrt{L}} \rfloor, \qquad P_z = \prod_{\substack{3 \leq q \leq z \\ q \in \mathbb{P}}} q,$$

$$V_0(z) = \frac{1}{2} \prod_{\substack{3 \leq q \leq z \\ q \in \mathbb{P}}} \left(1 - \frac{1}{q}\right), \tag{2.2}$$

$$\mathcal{B}_X(n) = \frac{\mathbf{1}_{n \in \mathcal{I}_X} \mathbf{1}_{(n+2, P_z)=1}}{V_0(z)}.$$

On $\mathcal{I}_X$ set

$$A_X(n) = \Lambda(n+2), \qquad \Delta_X(n) = A_X(n) - \mathcal{B}_X(n). \tag{2.3}$$

Both A_X and $\mathcal{B}_X$ are nonnegative; Δ_X is signed. All sums below are restricted to their stated subsets of $\mathcal{I}_X$. They may equivalently be viewed as sums over $\mathbb{N}$ after zero extension.

Define the prime-input set and its two masses by

$$\mathcal{P}_X = I_X \cap \mathbb{P}, \qquad Q_X = \sum_{p \in \mathcal{P}_X} \Lambda(p+2), \qquad B_*(X) = \sum_{p \in \mathcal{P}_X} \mathcal{B}_X(p). \tag{2.4}$$

For large X, every $p \in \mathcal{P}_X$ is odd. The comparison in (2.2) must not be replaced by a singular-series weight or by $\Lambda(n+2)$. Its cutoff tests only the absence of small prime divisors of $n+2$; it does not test whether $n+2$ is prime.

2.2 The fixed Ford weight

Put

$$\nu_0 = \frac{16623}{100000}, \qquad 0 < \varepsilon \le \frac{\nu_0}{100}, \qquad \gamma = \frac{1}{2} - \varepsilon, \qquad \sigma = \nu_0 - 2\varepsilon, \qquad \delta_g = \frac{1}{2} - 2\varepsilon. \tag{2.5}$$

For a vector $\mathbf{t}$, write $|\mathbf{t}|$ for the sum of its coordinates, not its dimension. The empty vector is denoted by $\varnothing$. The endpoint function g_0 is the symmetric function from the proof of Ford–Maynard Theorem 2.7(b), equation (8.5) [1], extended by zero:

$$g_0(\mathbf{t}) = \begin{cases} 1, & \mathbf{t} = \varnothing, \\ -1, & \dim \mathbf{t} \in \{1,3\},\ t_i > \nu_0\ (\forall i),\ |\mathbf{t}| \le 1/2, \\ 1, & \dim \mathbf{t} = 2,\ t_i > \nu_0\ (\forall i),\ (1-\nu_0)/2 < |\mathbf{t}| < 1/2, \\ 0, & \text{otherwise.} \end{cases} \tag{2.6}$$

Set $g_\varepsilon(\varnothing) = 1$ and, on nonempty vectors,

$$g_\varepsilon(\mathbf{t}) = g_0(\mathbf{t}) \mathbf{1}_{|\mathbf{t}| \le \delta_g}. \tag{2.7}$$

The separate truncation in (2.7) is retained. It is not interchangeable with the outer cutoff $uv \le n^\gamma$ below.

For $n > 1$, split its *complete prime-power factors* as

$$a(n) = \prod_{q^{v_q(n)} \| n,\ q < n^\sigma} q^{v_q(n)}, \qquad b(n) = \prod_{q^{v_q(n)} \| n,\ q \ge n^\sigma} q^{v_q(n)}. \tag{2.8}$$

Thus $a(n)b(n) = n$ and $(a(n), b(n)) = 1$. If $v = q_1 \cdots q_j$ is a factorization into primes with multiplicity, define

$$\mathbf{v}(v; n) = \left(\frac{\log q_1}{\log n}, \dots, \frac{\log q_j}{\log n} \right), \qquad \mathbf{v}(1; n) = \varnothing.$$

The function is symmetric, so the ordering of repeated coordinates is immaterial. The source weight is

$$H_\varepsilon(n) = \sum_{v \mid b(n)} g_\varepsilon(\mathbf{v}(v; n)) \sum_{\substack{u \mid a(n) \\ uv \le n^\gamma}} \mu(u). \tag{2.9}$$

This pins the shrunk version used in the current source records [6, 8]. It does not assert that every earlier unshrunk implementation is identical.

Lemma 2.1. *The weight in* (2.9) *satisfies* $|H_\varepsilon(n)| \le \tau(n)$ *for every* $n > 1$, *and* $H_\varepsilon(p) = 1$ *for every prime* p.

Proof. Each pair (u, v) in (2.9) determines a distinct divisor $uv \mid n$, because the prime supports of $a(n)$ and $b(n)$ are disjoint. Each coefficient has absolute value at most one. There are at most $\tau(n)$ divisors. If $n = p$ is prime, then $a(p) = 1$, $b(p) = p$, and $p > p^\gamma$. Only $(u, v) = (1, 1)$ contributes, with value one. □

2.3 The interior and exterior sets

For $P_\varepsilon = (\gamma, \varepsilon, \sigma)$, let $\mathcal{R}(P_\varepsilon)$ consist of all finite vectors $\mathbf{x} = (x_1, \dots, x_k)$ with

$$0 < x_i < 1 - \gamma, \qquad \sum_i x_i = 1, \qquad \sum_{i \in J} x_i \notin [\varepsilon, \varepsilon + \sigma] \quad (\varnothing \neq J \subsetneq \{1, \dots, k\}). \tag{2.10}$$

A *coagulation* of a vector is obtained by partitioning its coordinates into nonempty blocks and replacing each block by its sum. Let $\mathcal{C}(\mathcal{R}(P_\varepsilon))$ be the set of all such coagulations of vectors in $\mathcal{R}(P_\varepsilon)$. These are the definitions in [1, Definitions 4.3 and 4.5].

With prime factors counted with multiplicity in $\mathbf{v}(n; n)$, define

$$\begin{aligned} \mathcal{N}_{\varepsilon,X} = \{n \in \mathcal{I}_X : n \text{ is composite}, \\ \mathbf{v}(n; n) \in \mathcal{C}(\mathcal{R}(P_\varepsilon))\}, \\ \mathcal{E}_{\varepsilon,X} = \mathcal{I}_X \setminus (\mathcal{P}_X \cup \mathcal{N}_{\varepsilon,X}). \end{aligned} \tag{2.11}$$

In particular, the exterior is not an arbitrarily selected collection of favourable factorization regions.

The full physical discrepancy and the exterior discrepancy are, respectively,

$$T_X = \sum_{n \in \mathcal{I}_X} H_\varepsilon(n) \Delta_X(n), \qquad S_X = \sum_{n \in \mathcal{E}_{\varepsilon,X}} H_\varepsilon(n) \Delta_X(n). \tag{2.12}$$

The current direct project target concerns S_X, not T_X and not an isolated A5 component. Every H_ε and every set in this manuscript uses the same fixed ε.

2.4 Two explicit interior quantities

For a real number t, let $t_+ = \max(t, 0)$. Define

$$C_X = B_*(X) + \sum_{n \in \mathcal{N}_{\varepsilon,X}} H_\varepsilon(n) \mathcal{B}_X(n), \qquad D_X = \sum_{n \in \mathcal{N}_{\varepsilon,X}} (H_\varepsilon(n))_+ \Lambda(n+2). \tag{2.13}$$

Here $D_X \geq 0$ measures precisely the part of the interior weight that cannot be discarded by a nonpositivity argument. Using D_X makes the present endgame independent of an unspecified "good/bad" partition. It does not make D_X small by definition. Section 4.3 gives the exact conversion from the project's good/bad estimates to (2.13).

3 The precise conditional endgame

3.1 An exact identity and the sign of the exterior term

Lemma 3.1 (Finite partition identity)**.** *For the quantities in Section 2, one has the exact equality*

$$Q_X = C_X + T_X - S_X - \sum_{n \in \mathcal{N}_{\varepsilon,X}} H_\varepsilon(n) \Lambda(n+2). \tag{3.1}$$

Consequently,

$$\boxed{Q_X \geq C_X + T_X - S_X - D_X.} \tag{3.2}$$

Proof. Partition $\mathcal{I}_X$ into the three disjoint sets $\mathcal{P}_X$, $\mathcal{N}_{\varepsilon,X}$ and $\mathcal{E}_{\varepsilon,X}$. By Lemma 2.1,

$$\begin{aligned} T_X &= \sum_{p\in\mathcal{P}_X} \Delta_X(p) + \sum_{n\in\mathcal{N}_{\varepsilon,X}} H_\varepsilon(n)\Delta_X(n) + S_X \\ &= Q_X - B_*(X) + \sum_{n\in\mathcal{N}_{\varepsilon,X}} H_\varepsilon(n)\big(\Lambda(n+2) - \mathcal{B}_X(n)\big) + S_X. \end{aligned}$$

Solving this equality for Q_X gives (3.1). Since $\Lambda(n+2) \geq 0$,

$$\sum_{n\in\mathcal{N}_{\varepsilon,X}} H_\varepsilon(n)\Lambda(n+2) \leq \sum_{n\in\mathcal{N}_{\varepsilon,X}} (H_\varepsilon(n))_+\Lambda(n+2) = D_X.$$

This proves (3.2). □

Remark 3.2 (Sign convention)**.** For the explicit definitions above, the exterior term occurs with a *minus* sign. Therefore an upper bound on S_X suffices, and negative values of S_X help. No absolute bound on S_X is required. This sign is also visible in the partition used in [1, p. 54]. An older abstract dossier [11] displayed an endgame with a differently signed error. That display is not imported here: (3.1) is the source-level identity used in this note, and any proposed correspondence to an older error variable must preserve it.

3.2 All hypotheses needed for the direct deduction

Hypothesis 3.3 (Same-source endgame inputs)**.** There exist fixed constants $c_F > 0$, $b_0 > 0$, a fixed admissible $\varepsilon > 0$, and a threshold X_0, such that, for every $X \geq X_0$, the source in Section 2 satisfies

$$B_*(X) \geq b_0 \frac{X}{L}, \qquad \text{comparison mass,} \tag{3.3}$$

$$|T_X| \leq \frac{c_F}{16} B_*(X), \qquad \text{complete physical Type-I error,} \tag{3.4}$$

$$C_X \geq \frac{7c_F}{8} B_*(X), \qquad \text{interior comparison positivity,} \tag{3.5}$$

$$D_X \leq \frac{c_F}{16} B_*(X), \qquad \text{positive interior leakage.} \tag{3.6}$$

The same H_ε, $\mathcal{N}_{\varepsilon,X}$, $\mathcal{B}_X$ and $B_*(X)$ must occur in all four inequalities. These are explicit prerequisites, not unnamed assumptions encapsulated in a software predicate.

The fractions in this hypothesis are one convenient budget. More generally, (3.2) gives a positive detector whenever the lower budget for C_X exceeds the sum of the budgets for $|T_X|$, D_X and S_X.

Theorem 3.4 (One-sided exterior control implies twin-prime abundance)**.** *Assume Hypothesis 3.3. Suppose that for some fixed $\rho \geq 0$ and all sufficiently large X,*

$$S_X \leq \left(\frac{c_F}{4} - \rho\right) B_*(X). \tag{3.7}$$

Then, for every sufficiently large X,

$$Q_X \geq \left(\frac{c_F}{2} + \rho\right) B_*(X) \geq \frac{c_F b_0}{2} \frac{X}{L}. \tag{3.8}$$

Moreover,

$$T_2(X) := \#\{p \in (X/2, X] : p, p+2 \in \mathbb{P}\} \geq \frac{c_F b_0}{8} \frac{X}{L^2}. \tag{3.9}$$

In particular, there are infinitely many twin primes.

Proof. Insert (3.3)–(3.6) and (3.7) into (3.2). The coefficient of $c_F B_*(X)$ is

$$\frac{7}{8} - \frac{1}{16} - \frac{1}{4} - \frac{1}{16} = \frac{1}{2}.$$

This proves (3.8).

It remains to distinguish primes from proper prime powers in the detector. For $Z \geq 4$, the number of integers of the form $q^a \leq Z$, where q is prime and $a \geq 2$, is at most

$$\sum_{a=2}^{\lfloor \log_2 Z \rfloor} \lfloor Z^{1/a} \rfloor \leq Z^{1/2} \log_2 Z.$$

Repeated representations only increase this upper bound. Since $\Lambda(q^a) \leq \log Z$, the total contribution to Q_X from $p+2$ that are proper prime powers is at most

$$O\big(X^{1/2}(\log X)^2\big) = o(X/L). \tag{3.10}$$

For genuine twin pairs, $\Lambda(p+2) = \log(p+2) \leq \log(X+2) \leq 2L$ when X is sufficiently large. Therefore

$$Q_X \leq 2L\, T_2(X) + O\big(X^{1/2}(\log X)^2\big).$$

Choose X large enough that the final error is at most $c_F b_0 X/(4L)$, and use (3.8). Then

$$2L\, T_2(X) \geq \frac{c_F b_0}{4} \frac{X}{L},$$

which proves (3.9). Given any bound M, choose such an $X > 2M$. The lower bound is positive, so there is a twin pair with its smaller member above M. Thus the set of twin pairs is infinite. □

Status: Theorem 3.4 is a proved implication. Its open exterior hypothesis (3.7) has not been proved in this manuscript or in the latest supplied report. The supporting source estimates must also be verified at the precise scope stated above.

3.3 Quantifiers and a sufficient choice of ε

The source records propose stronger asymptotic forms of the supporting inputs:

$$\begin{aligned} B_*(X) &= (C_2 + o(1))X/L, \qquad T_X = o_\varepsilon(B_*(X)), \\ C_X/B_*(X) &\geq c_F - K_G\varepsilon - o_\varepsilon(1), \qquad D_X/B_*(X) \leq K_D\varepsilon + o_\varepsilon(1), \end{aligned} \tag{3.11}$$

where $C_2 > 0$ and K_G, K_D are independent of sufficiently small positive ε. If these assertions hold, first choose one ε for which $K_G\varepsilon \leq c_F/16$ and $K_D\varepsilon \leq c_F/32$. Next take X large enough that the corresponding $o_\varepsilon(1)$ terms are at most $c_F/16$ and $c_F/32$, respectively, and $|T_X|/B_*(X) \leq c_F/16$. Taking, for example, $b_0 = C_2/2$ gives Hypothesis 3.3.

This argument is existential and asymptotic. A small printed value of ε alone does not verify unknown constants or give an effective X_0.

4 The supporting estimates: provenance and exact verification tasks

The following discussion identifies why the supporting inputs are distinct from the hard exterior correlation. It records the mathematical route in the supplied sources, not a claim that every source-specific analytic detail has been independently re-proved during preparation of this note.

4.1 The comparison is a rough shifted-prime count, not a twin-prime count

Let

$$W_0(z) = \prod_{3\le q\le z}\left(1-\frac{1}{q-1}\right), \qquad C_2 = \prod_{q>2}\left(1-\frac{1}{(q-1)^2}\right).$$

The exact identity

$$\frac{W_0(z)}{V_0(z)} = 2\prod_{3\le q\le z}\left(1-\frac{1}{(q-1)^2}\right) \tag{4.1}$$

follows by simplifying each Euler factor. The infinite product C_2 is positive: all factors are positive, and $\sum_{q>2}(q-1)^{-2} < \infty$.

The comparison-mass assertion used in [5, Section 4] is the sieve asymptotic

$$\#\{p \in I_X \cap \mathbb{P} : (p+2, P_z) = 1\} = (1+o(1))\, W_0(z)\frac{X}{2L}. \tag{4.2}$$

Together with (4.1), it yields $B_*(X) = (C_2+o(1))X/L$. The input to (4.2) is prime distribution in progressions at a fixed level below $X^{1/2}$ and the fundamental lemma of the sieve; $\log z = \sqrt{L}$ and the sieve parameter tends to infinity. The relevant classical tools are described in [2, 3, 4].

The need for this exact normalization should not be hidden behind a generic "comparison-mass" statement. In particular, (4.2) does not assert that $p+2$ is prime: factors larger than z remain possible. The detailed source-dependent comparison proof is recorded in [5]; its uniform prime-prefix version is a separate item for review.

4.2 The full physical sum is a short-divisor expression

Each divisor $d \mid n$ has a unique decomposition $d = d_1d_2$ according to (2.8). Define

$$G_\varepsilon(d;n) = \mathbf{1}_{d\le n^\gamma}\mu(d_1)g_\varepsilon(\mathbf{v}(d_2);n)). \tag{4.3}$$

Then

$$H_\varepsilon(n) = \sum_{d\mid n} G_\varepsilon(d;n), \qquad G_\varepsilon(d;n) = 0 \quad (d > n^\gamma). \tag{4.4}$$

The project Type-I argument [8, 9] requires, and records, a bounded-variation estimate

$$\|G_\varepsilon(d;\cdot)\|_\infty + \mathrm{Var}_{I_X}\, G_\varepsilon(d;\cdot) \ll_\varepsilon L \tag{4.5}$$

with the literal cutoffs and zero extension retained. The associated source-specific progression lemma is

$$\sum_{n\in\mathcal{I}_X}\Delta_X(n)\sum_{d\mid n}c_d(n) \ll_{A,B,C,\delta} XL^{-A}, \tag{4.6}$$

when $d \le X^{1/2-\delta}$ is odd and $\|c_d\|_\infty + \mathrm{Var}\, c_d \le L^B\tau(d)^C$, for fixed $A, B, C, \delta > 0$. For the prime part, $d \mid n$, n odd implies $n+2 \equiv d+2 \pmod{2d}$, a reduced class for odd d. The comparison part must be evaluated using (2.2), with matching local main terms. Divisor-weight insertion, partial summation, and their total error are part of (4.6), not consequences of merely naming Bombieri–Vinogradov.

If (4.5) and (4.6) are established for the specified source, (4.4), with $\delta = \varepsilon$, gives $T_X \ll_{A,\varepsilon} XL^{-A}$. In particular $T_X = o_\varepsilon(B_*(X))$ for $A > 1$. There is no arbitrary exterior mask in this use of (4.6).

4.3 Geometric positivity and the original collar

The source records use a partition $\mathcal{N}_{\varepsilon,X} = \mathcal{N}_{1,\varepsilon,X} \sqcup \mathcal{N}_{2,\varepsilon,X}$ with $H_\varepsilon \le 0$ on $\mathcal{N}_1$. The second set is the exceptional geometric collar; no part of it is silently removed from S_X.

Here is a precise sufficient package from which the interior inputs follow. Suppose, for one and the same partition, that

$$B_*(X) + \sum_{\mathcal{N}_1} H_\varepsilon(n)\mathcal{B}_X(n) \ge (c_F - K_1\varepsilon - o_\varepsilon(1))B_*(X), \tag{4.7}$$

$$\sum_{\mathcal{N}_2} |H_\varepsilon(n)|\mathcal{B}_X(n) \le (K_2\varepsilon + o_\varepsilon(1))B_*(X), \tag{4.8}$$

$$\sum_{\mathcal{N}_2} |H_\varepsilon(n)|\Lambda(n+2) \le (K_3\varepsilon + o_\varepsilon(1))B_*(X). \tag{4.9}$$

All constants are to be uniform for sufficiently small fixed ε. Then, by adding back the collar in the comparison sum,

$$C_X \ge (c_F - (K_1 + K_2)\varepsilon - o_\varepsilon(1))B_*(X).$$

Also $(H_\varepsilon)_+ = 0$ on $\mathcal{N}_1$, so

$$D_X \le \sum_{\mathcal{N}_2} |H_\varepsilon(n)|\Lambda(n+2) \le (K_3\varepsilon + o_\varepsilon(1))B_*(X).$$

Thus (4.7)–(4.9) imply the last two assertions in (3.11). This conversion is an elementary deduction; it explains precisely the relationship between the all-interior formulation here and the earlier good/bad formulation.

The proposed justification in [5, Sections 4, 6–7] is as follows. The nonzero collar has boundedly many prime factors, all of fixed positive-power size. A prime-vector comparison theorem gives (4.8) from the $O(\varepsilon)$ volume of the collar. For (4.9), write a squarefree input as $n = Mq$, with q its largest prime factor. A two-dimensional *upper* sieve is applied to $q(Mq+2)$. It gives a bound of order $(X/M)/L^2$ on each dyadic q range. The collar constraint is charged to the prefix reciprocal measure $\sum_M 1/M = O(\varepsilon) + o_\varepsilon(1)$, rather than to an unproved prime theorem in very short intervals. Multiplication by the output logarithm gives the required scale $\varepsilon X/L$. Repeated factors and proper prime-power outputs must be removed separately.

These are upper-bound and comparison tasks, not the unproved positive exterior correlation. Nevertheless, verifying the actual collar, all its proper-subsum boundaries, the last-prime uniformity, and the same-source normalization is necessary. This note does not replace that verification by the assertion that a thin set is automatically negligible.

4.4 The limiting constant and its evidence boundary

For $2 \le k \le 6$, set

$$\Sigma_k = \{(x_1, \dots, x_k) : x_i \ge \nu_0,\ \textstyle\sum_i x_i = 1\}, \qquad h_0(\mathbf{x}) = \textstyle\sum_{J\subseteq\{1,\dots,k\}} g_0(\mathbf{x}_J).$$

Use the projection measure $dx_1 \cdots dx_{k-1}$ with $x_k = 1 - \sum_{i<k} x_i$. The limiting constant in the project is

$$c_F = 1 + \sum_{k=2}^{6} \frac{1}{k!} \int_{\Sigma_k} \frac{h_0(\mathbf{x})}{x_1 \cdots x_k}\, dx_1 \cdots dx_{k-1}. \tag{4.10}$$

The factor $1/k!$ is essential because the simplex in this definition is unordered. On an ordered simplex it is omitted. Boundaries have zero projection measure; the discrete source still uses the exact endpoint conventions in (2.6)–(2.9).

The public endpoint calculation in [1, proof of Theorem 2.7(b), pp. 65–67] reports a lower bound $6 \cdot 10^{-6}$ for this constant. The supplied rational integration report [7] reports the stronger enclosure

$$0.0000068000577158 < c_F < 0.0000068000577163. \tag{4.11}$$

The latter computation was *not rerun* in preparing this note. Neither its extra decimal places nor an effective onset is needed for Theorem 3.4; a rigorously justified positive lower bound suffices. A referee must distinguish the finite positivity certificate from the arithmetic comparison and collar estimates used to transfer it to C_X.

5 The current rough-input and moment hypotheses

This section states, without upgrading their verification status, the two project-specific inputs needed to use the latest degree-twelve reduction. They are stronger than the elementary implications proved in the next section.

5.1 The complete exterior roughening statement

For the same fixed ε, define

$$K_X = \lceil 448 \log L \rceil, \qquad Y = X^{\varepsilon/(8K_X)}, \qquad \theta = \frac{\log Y}{L} = \frac{\varepsilon}{8K_X}, \tag{5.1}$$

and

$$\mathcal{Q}_Y = \{n \in \mathcal{I}_X : \mu(n)^2 = 1,\ P^-(n) > Y\}, \qquad S_Y = \sum_{n \in \mathcal{E}_{\varepsilon,X} \cap \mathcal{Q}_Y} H_\varepsilon(n) \Delta_X(n). \tag{5.2}$$

Here $P^-(n)$ is the least prime factor of n.

Hypothesis 5.1 (Rough-input transfer)**.** For the complete exterior source, and not merely for a selected factorization region,

$$S_X = S_Y + O_\varepsilon(XL^{-4}). \tag{5.3}$$

The supporting project proofs are [8, 9]. Their mechanism is a finite roughness expansion coupled with (4.4): the combined modulus remains strictly below $X^{1/2}$, and squareful inputs are removed with a separately estimated error. The passage from the full physical sum to the exterior uses vanishing on its prime/interior complement; it is not an application of Type I to an arbitrary mask. This is a source-specific analytic deduction to verify, not a general principle that smaller supports have smaller signed sums.

For later use, elementary calculus gives, with constants depending on fixed ε,

$$\theta \asymp_\varepsilon (\log L)^{-1}, \qquad \log Y \asymp_\varepsilon \frac{L}{\log L}. \tag{5.4}$$

Hence Y eventually exceeds e^{L^a} for every fixed $a < 1$, but $Y = X^{o(1)}$. These statements do not hold uniformly as $\varepsilon \to 0$.

5.2 One translated count and its correct local mean

Define

$$y = e^{L^{1/10}}, \qquad w = e^{L^{1/5}}, \qquad \mathscr{P}_X = \{p \in \mathbb{P} : y < p \le w\}, \tag{5.5}$$

$$R(n) = \sum_{p \in \mathscr{P}_X} \mathbf{1}_{p \mid n+1}, \qquad V = \sum_{p \in \mathscr{P}_X} \frac{1}{p-2}. \tag{5.6}$$

For large X, $\mathscr{P}_X$ is nonempty and $V > 0$. Mertens' reciprocal-prime estimate, and $1/(p-2) - 1/p = O(p^{-2})$, give

$$V = \frac{1}{10}\log L + o(1). \tag{5.7}$$

The letter w in (5.5) is a cutoff, not the normalized discrepancy sequence introduced in Section 7.

Lemma 5.2 (Finite local calculation)**.** *Let $p > 2$ be prime and $p \nmid c$. Among residues $q \pmod p$ excluding $q = 0$ and $cq = -2$, the proportion satisfying $cq = -1$ is $1/(p-2)$.*

Proof. Since c is invertible modulo p, the forbidden residues 0 and $-2c^{-1}$ are distinct. The residue $-c^{-1}$ is distinct from both. Exactly one of the $p-2$ allowed residues has the required property. □

This is a finite residue calculation. It does not state independence of the primality conditions $q \in \mathbb{P}$ and $cq + 2 \in \mathbb{P}$.

Hypothesis 5.3 (Uniform twelfth-moment bound)**.** For every fixed $c_0 > 0$ and $A > 0$, uniformly for $c_0/\log L \le \theta \le 1/4$, with $Y = X^\theta$, one has

$$\sum_{n \in \mathcal{Q}_Y} \tau(n)\big(\Lambda(n+2) + \mathcal{B}_X(n)\big)|R(n) - V|^{12} \ll_{c_0,A} \frac{XV^6}{L\theta^4} + XL^{-A}. \tag{5.8}$$

The constants and threshold may depend on c_0, A, but not on θ in the displayed range.

The report [10, Theorem 1] proposes the corresponding $2k$-th moment for every fixed k. This note uses only $k = 6$. The report supplies a paper-level argument with upper sieves, finite residue moments and lattice errors; no independent complete analytic certification of that argument is claimed here.

5.3 What does and does not prove the moment hypothesis

Some constituent calculations can be stated and checked independently. If $n = cq \in \mathcal{Q}_Y$ and $q = P^+(n)$, then $\tau(n) = 2\tau(c)$ and

$$\sum_{\substack{c \le X/Y \\ \mu(c)^2 = 1,\ P^-(c) > Y}} \frac{2\tau(c)}{c} \le 2 \prod_{Y < p \le X} \left(1 + \frac{2}{p}\right) \ll \theta^{-2}. \tag{5.9}$$

The inequality follows by expanding the nonnegative Euler product. The final bound follows from $\sum_{Y<p\le X} 1/p = \log(1/\theta) + O(1)$ and $\sum p^{-2} < \infty$.

In a *complete finite CRT model*, condition on excluding the two bad residues at every auxiliary prime. The indicators in Lemma 5.2 are then independent Bernoulli variables B_p of means $v_p = 1/(p-2)$. Expanding $\mathbb{E}(\sum_p (B_p - v_p))^{12}$, any index appearing exactly once has zero expectation. All surviving terms involve at most six distinct primes. For an exponent

$a \geq 2$, $\mathbb{E}|B_p - v_p|^a \leq \mathbb{E}|B_p - v_p|^2 \leq v_p$. Summing over the finitely many set-partition types therefore gives

$$\mathbb{E}\left(\sum_p (B_p - v_p)\right)^{12} \ll (1+V)^6. \tag{5.10}$$

Equations (5.9) and (5.10) do *not* by themselves prove (5.8). The remaining analytic passage must bound actual sifted integers and actual prime outputs using a nonnegative majorant, preserve the centered main term before taking absolute values, and control the total remainder over all prefixes. The report proposes $D = Y^{1/10}$ and a total rounding bound

$$X \exp\left(-\frac{4}{5}\log Y + O_A(\sqrt{L}\log L)\right) L^{O_{c_0}(1)}. \tag{5.11}$$

Given such a bound, (5.4) makes it $O(XL^{-A'})$ for any fixed $A' > 0$; the asserted bound itself, its Selberg normalization, and the literal comparison's different sieve dimension are the matters requiring review. No assertion about genuine-prime independence is a substitute for this step.

6 A complete deduction of the degree-twelve reduction

6.1 Restoration error

Define the fixed-degree weight

$$M_{12}(n) = 1 - \left(1 - \frac{R(n)}{V}\right)^{12}, \qquad S_{12}(X) = \sum_{n \in \mathcal{E}_{\varepsilon,X} \cap \mathcal{Q}_Y} H_\varepsilon(n)\Delta_X(n)M_{12}(n). \tag{6.1}$$

This is a real signed weight. It equals zero if $R(n) = 0$, but can be negative when $R(n) > 2V$. No probability interpretation is used.

Proposition 6.1 (Conditional full-source restoration)**.** *Assume Hypotheses 5.1 and 5.3. For the cutoff* (5.1) *and fixed $\varepsilon > 0$,*

$$S_X = S_{12}(X) + O_\varepsilon\left(\frac{X}{L(\log L)^2}\right). \tag{6.2}$$

If (3.3) *holds, the error is $o(B_*(X))$.*

Proof. The exact expression for the difference is

$$S_Y - S_{12} = \sum_{n \in \mathcal{E}_{\varepsilon,X} \cap \mathcal{Q}_Y} H_\varepsilon(n)\Delta_X(n)\left(1 - \frac{R(n)}{V}\right)^{12}.$$

Use Lemma 2.1 and $|\Delta_X(n)| \leq \Lambda(n+2) + \mathcal{B}_X(n)$ to obtain

$$\begin{aligned} |S_Y - S_{12}| &\leq V^{-12} \sum_{n \in \mathcal{Q}_Y} \tau(n)[\Lambda(n+2) + \mathcal{B}_X(n)]|R(n) - V|^{12} \\ &\ll \frac{X}{L\theta^4 V^6} + XL^{-4}V^{-12}, \end{aligned}$$

where (5.8) is used with $A = 4$ and a fixed c_0 small enough for (5.1). Equations (5.4) and (5.7) imply $\theta^{-4} \ll_\varepsilon (\log L)^4$ and $V^{-6} \ll (\log L)^{-6}$. The first term is therefore $O_\varepsilon(X/[L(\log L)^2])$. For large X, $V \geq 1$, and the second term is absorbed by the same bound. Finally, (5.3) contributes $O_\varepsilon(XL^{-4})$, which is also absorbed. Dividing by the lower bound in (3.3) proves the last assertion. □

This proposition pays a restoration error; it provides no upper bound for S_{12} itself. In particular, it remains true for suitable nonnegative comparison controls with nonzero main mass.

6.2 An explicit divisor sum, without integral or no-hit masks

For $1 \le \ell \le 12$, put

$$a_\ell(V) = \ell! \sum_{j=\ell}^{12} (-1)^{j+1} \binom{12}{j} \left\{ {j \atop \ell} \right\} V^{-j}, \tag{6.3}$$

where $\left\{ {j \atop \ell} \right\}$ is a Stirling number of the second kind. Let

$$\mathscr{D}_X = \{d \ge 2 : \mu(d)^2 = 1,\ p \mid d \Rightarrow p \in \mathscr{P}_X,\ 1 \le \omega(d) \le 12\}. \tag{6.4}$$

Lemma 6.2 (Exact polynomial expansion)**.** *For every positive integer n,*

$$M_{12}(n) = \sum_{\substack{d \in \mathscr{D}_X \\ d \mid n+1}} a_{\omega(d)}(V). \tag{6.5}$$

Proof. Expanding the binomial gives $M_{12} = \sum_{j=1}^{12} (-1)^{j+1} \binom{12}{j} R^j V^{-j}$. For a nonnegative integer R, $R^j = \sum_{\ell=1}^{j} \left\{ {j \atop \ell} \right\} \ell! \binom{R}{\ell}$. The number $\binom{R(n)}{\ell}$ counts the squarefree divisors of $n+1$ formed from exactly ℓ distinct primes in $\mathscr{P}_X$. Substituting this identity and interchanging finite sums proves the formula. □

It follows, with no approximation, that $S_{12}(X) = \mathcal{J}_{12}(X)$, where

$$\boxed{\mathcal{J}_{12}(X) = \sum_{d \in \mathscr{D}_X} a_{\omega(d)}(V) \sum_{\substack{n \in \mathcal{E}_{\varepsilon,X} \cap \mathcal{Q}_Y \\ n \equiv -1 \pmod d}} H_\varepsilon(n)\, [\Lambda(n+2) - \mathcal{B}_X(n)].} \tag{6.6}$$

All physical endpoints, input roughness, squarefreeness, exterior restrictions, and both sources of signs are present in (6.6).

Lemma 6.3 (Size and harmonic cost of the auxiliary divisors)**.** *Every $d \in \mathscr{D}_X$ satisfies $d \le e^{12L^{1/5}}$. For sufficiently large X,*

$$\sum_{d \in \mathscr{D}_X} \frac{|a_{\omega(d)}(V)|}{d} \ll 1, \qquad \sum_{d \in \mathscr{D}_X} \frac{|a_{\omega(d)}(V)|}{\varphi(d)} \ll 1. \tag{6.7}$$

The constants are absolute once the degree twelve is fixed.

Proof. The bound on d follows from at most twelve factors, each at most w. For $V \ge 1$, the finite formula (6.3) gives $|a_\ell(V)| \le C_\ell V^{-\ell}$. The elementary symmetric-sum inequality implies

$$\sum_{\substack{d \in \mathscr{D}_X \\ \omega(d) = \ell}} \frac{1}{d} \le \frac{1}{\ell!} \left(\sum_{p \in \mathscr{P}_X} \frac{1}{p} \right)^\ell.$$

The prime sum is $V + o(1)$. The identical argument for $1/\varphi(d)$ uses $\sum_{p \in \mathscr{P}_X} 1/(p-1) = V + o(1)$. Sum over $1 \le \ell \le 12$. □

These harmonic bounds do not say that the ordinary number of d's is polylogarithmic. Nor do they make the inner arithmetic discrepancy small. If a native factorization $n = qm$ is opened, the exact conditions become

$$n + 1 = d\ell, \qquad d\ell - qm = 1, \qquad n + 2 = d\ell + 1, \qquad (d, qm) = 1. \tag{6.8}$$

They are a description of the remaining source, not a distribution theorem.

6.3 The precise open assertion and its consequence

Hypothesis 6.4 (Outstanding degree-twelve signed estimate)**.** For the fixed ε used in the endgame,

$$\limsup_{X\to\infty} \frac{\mathcal{J}_{12}(X)}{B_*(X)} \le \frac{c_F}{8}. \tag{6.9}$$

Status: Hypothesis 6.4 is not proved. It is the first remaining signed prime-output estimate in the latest degree-twelve formulation. The moment bound, a finite divisor expansion, bounded harmonic mass, or centering row means does not imply it.

Corollary 6.5 (Conditional degree-twelve route to twin primes)**.** *Assume Hypothesis 3.3 and both transfer hypotheses 5.1 and 5.3. If Hypothesis 6.4 also holds, then* (3.7) *holds eventually with* $\rho = c_F/16$*, and* (3.9) *follows.*

Proof. By (6.9), eventually $\mathcal{J}_{12}(X) \le (c_F/8 + c_F/32)B_*(X) = 5c_F B_*(X)/32$. By Proposition 6.1, eventually $|S_X - \mathcal{J}_{12}(X)| \le c_F B_*(X)/32$. Consequently

$$S_X \le \frac{6c_F}{32} B_*(X) = \frac{3c_F}{16} B_*(X) = \left(\frac{c_F}{4} - \frac{c_F}{16}\right) B_*(X).$$

Apply Theorem 3.4. □

The displayed ρ is a *conditional budget calculation.* It is not an achieved numerical reserve. A different complete-source bound with sufficient slack could replace (6.9); this particular hypothesis is sufficient, not necessary.

7 An alternative R11/Gram route, and its exact additional burden

This section explains what an R11 theorem would have to supply if the direct bound (3.7) is approached through physical bilinear rows. It is not used in Corollary 6.5.

7.1 Normalization

With the convention of the older normalized sources [5, 11],

$$a_X(n) = \frac{\Lambda(n+2)}{L}, \quad b_X(n) = \frac{\mathcal{B}_X(n)}{L}, \quad w_X(n) = \frac{\Delta_X(n)}{L}, \quad B_X = \frac{B_*(X)}{L}, \quad E_H(X) = \frac{S_X}{L}, \tag{7.1}$$

on the odd physical support. Hence a Gram formed with two copies of w_X has an L^{-2} normalization relative to the one formed with Δ_X. This factor must not be lost.

7.2 A sufficient conditional physical provider

Suppose a proved, source-exact reduction gives, for each requested $A > 0$,

$$|E_H(X)| \le K_A L^{d_A} \sup_{\omega \in \Omega_A(X)} |U_\omega(X)| + K_A X L^{-A-2} + K_A X^{1-\eta_A} L^{d_A}, \tag{7.2}$$

where $\eta_A > 0$ and the entire generated family $\Omega_A(X)$ is specified. Suppose its rows have the exact form

$$U_\omega = \sum_{c \asymp C_\omega} \Gamma_\omega(c) \sum_{m \asymp M_\omega} A_\omega(m) W_\omega(m,c) w_X(mc), \qquad M_\omega C_\omega \asymp X.$$

The profiles must retain the original physical mask, twists, tuple multiplicities and companion terms. Assume, uniformly over this entire family,

$$\sum_c |\Gamma_\omega(c)|^2 \ll C_\omega L^{J_A}, \tag{7.3}$$

$$\mathcal{G}_\omega := \sum_c \left| \sum_m A_\omega(m) W_\omega(m,c) w_X(mc) \right|^2 \ll_{A,Q} X M_\omega L^{-Q} \quad \text{for every fixed } Q > 0. \tag{7.4}$$

Proposition 7.1 (Conditional Gram export)**.** *Under* (7.2)–(7.4)*, $E_H(X) \ll_A X L^{-A}$ for every fixed $A > 0$. Consequently $S_X = o(B_*(X))$ whenever* (3.3) *holds.*

Proof. Cauchy–Schwarz in c gives

$$|U_\omega|^2 \le \left(\sum_c |\Gamma_\omega(c)|^2 \right) \mathcal{G}_\omega \ll X^2 L^{J_A - Q},$$

because $C_\omega M_\omega \asymp X$. Choose $Q = 2A + 2d_A + J_A + 4$ in (7.2). Its first two terms are $O(XL^{-A-2})$; its third is $o(XL^{-A})$. This proves the first claim. Since $S_X = LE_H(X)$, take $A > 2$ and divide by $B_*(X) \gg X/L$. □

Both the complete-source reduction (7.2) and the required family-wide bound (7.4) are real obligations. A theorem on one balanced packet, one modulus, one profile or one set of leaves does not establish them. Ford–Maynard Proposition 7.22 assumes Type II in proving the exterior estimate; it cannot be invoked backwards to manufacture the missing Type-II input.

An exact reconstruction $E_H = e_{\text{paid}} + \sum_\omega c_\omega K_\omega$ with a separately proved total coefficient budget could replace (7.2). Counting a finite number of combinatorial labels is not a bound on that budget. None of the current local or degree-twelve reductions is asserted here to instantiate the universal Gram hypothesis.

8 A referee's obligation ledger

Object	Exact role	Status in this note
Source definitions	Equations (2.2)–(2.11); same shrunk weight and same exterior throughout	Explicitly pinned. Any different historical version requires an additional identity.
Finite endgame	Lemma 3.1; comparison + Type-I error − exterior error − positive leakage	Proved here, without an analytic estimate.
Comparison prime mass	(3.3), from the exact rough comparison	Classical sieve route recorded in [5]; source normalization must be checked.
Complete physical Type I	(3.4), via (4.5)–(4.6)	Project analytic deduction based on classical BV; no independent full audit asserted here.
Geometric constant	$c_F > 0$ and the definition (4.10)	Published endpoint calculation and reported project rational certificate. Extra decimal enclosure not rerun.
Interior comparison and collar	(3.5), (3.6), or the sufficient package (4.7)–(4.9)	Project-specific comparison/upper-sieve arguments; exact scope and uniformity require review.
Rough-input transfer	Hypothesis 5.1	Written project deduction. Needed only for the degree-twelve route.
Twelfth moment	Hypothesis 5.3	Proposed analytic theorem with a paper-level argument; not independently certified here.
Moment-to-marker transfer	Proposition 6.1	Proved here conditional on the two displayed transfer inputs.
Degree-twelve algebra	Lemmas 6.2 and 6.3	Exact algebra and harmonic accounting; not prime-output cancellation.
Signed exterior estimate	(3.7), or (6.9) with the verified transfers	**Open.** No theorem in this note proves it.
Prime-power removal	(3.10) and Theorem 3.4	Proved here by elementary counting.
Full physical R11 export	(7.2)–(7.4)	An alternative sufficient route; not proved and not required by the direct route.
Lean / numerical onset	Independent kernel build; effective X_0	Neither performed nor claimed.

The conditional proof is complete as an implication, but the antecedent is not established in its entirety. "Only the final estimate remains" is justified *only after* every supporting source-specific estimate used by the chosen route has been verified. A paper-level claim of such a verification is not the verification itself.

For external review, the first priorities are the same-source definitions, the normalization and sign in Lemma 3.1, the comparison/collar interface, and the exact scope of the two degree-twelve transfer hypotheses. Only then should a proposed proof of (6.9) be assessed as a possible closing argument. No numerical census of A5 boxes substitutes for any of these analytic statements.

9 What the note establishes and what it does not

The note establishes the explicit implication in Theorem 3.4 and the further conditional implication in Corollary 6.5. The central chain is

$$\begin{gathered}\text{verified same-source supporting estimates}\\ +\ \text{rough-input transfer}\\ +\ \text{uniform twelfth moment}\\ +\ \text{one-sided bound for } \mathcal{J}_{12}\\ \Longrightarrow\ S_X \le 3c_F B_*(X)/16 \Longrightarrow\ Q_X \gg X/L \Longrightarrow\ T_2(X) \gg X/L^2.\end{gathered} \tag{9.1}$$

The direct route starts instead from the bound on S_X and does not require the two auxiliary transfer hypotheses. The Gram route is another sufficient way of supplying that bound if its full source contract is proved.

No argument here bounds the actual signed sum $\mathcal{J}_{12}$ at the required scale. The use of a small auxiliary modulus, a correct transform, or a normalized marker does not fill that gap. The final statement is therefore not an unconditional resolution of the Twin Prime Conjecture.

A Source and normalization checks that must not be omitted

A.1 Even inputs

The manuscript uses the odd source explicitly. If a source theorem was stated on all of I_X, its even part must be dealt with before identifying it with (2.12). Here $\mathcal{B}_X(n) = 0$ on even n. Also $\Lambda(n+2)$ can be nonzero for even n only if $n+2$ is a power of two. There are $O(L)$ such inputs. Using $|H_\varepsilon(n)| \le \tau(n) \ll_\delta X^\delta$, their total is $O_\delta(X^\delta L)$, negligible relative to X/L for any fixed $0 < \delta < 1$. This observation does not authorize deleting other components of a different source decomposition.

A.2 Repeated prime factors and unordered integrals

The arithmetic definition (2.9) includes repeated factors. The optional roughening hypothesis is what removes them from the retained source. A squarefree auxiliary divisor does not by itself make the input squarefree. The geometric integral in (4.10) is symmetric and unordered; an A5 box integral using ordered prime factors has a different counting convention. Those two integrals cannot be substituted for each other.

A.3 Direct scalar versus universal physical theorem

An error $O(B_*(X)/(\log\log X)^2)$ is sufficient for a fixed positive direct margin. It is not an arbitrary power of $\log X$ saving and therefore is not, without further work, a universal Gram provider. Conversely, the direct one-sided theorem does not need a stronger absolute Gram theorem as an extra hypothesis.

A.4 The bounded-class normalization issue

The normalized comparison $b_X = \mathcal{B}_X/L$ has prime mass of order X/L^2. The bounded-class positivity theorem discussed in [1] uses a different prime-mass hypothesis. A direct substitution

into that headline theorem is not licensed merely by boundedness of a_X, b_X. The finite direct endgame in Section 3 avoids that substitution, but retains the sparse comparison and collar estimates that must actually be verified.

B The reported finite geometric certificate

For clarity, the exact integrals used in the project positivity calculation can be specified independently of the decimal report. Put $h = (1-\nu_0)/2$ and define

$$\mathscr{I}_k = \int_{\Sigma_k} \frac{d\mathbf{x}}{x_1 \cdots x_k}, \qquad \mathscr{I}_{k,r}(a) = \int_{\Sigma_k} \frac{\mathbf{1}_{x_1+\cdots+x_r \le a}}{x_1 \cdots x_k}\, d\mathbf{x},$$

with the projection measure used in (4.10). The project report [7] expresses the same constant as

$$\begin{aligned} c_F &= 1 + J_3 + J_4 + J_5 + J_6, \\ J_3 &= \frac{2}{3}\mathscr{I}_3 - \mathscr{I}_{3,1}(1/2) - \frac{1}{2}\mathscr{I}_{3,2}(h), \\ J_4 &= -\frac{1}{4}\mathscr{I}_{4,2}(h), \\ J_5 &= -\frac{7}{60}\mathscr{I}_5 - \frac{1}{12}\mathscr{I}_{5,2}(h) + \frac{1}{6}\mathscr{I}_{5,2}(1/2), \qquad J_6 = -\frac{1}{48}\mathscr{I}_6. \end{aligned}$$

These are not additional assumptions in the finite endgame; they describe a way to verify its positive geometric constant.

The report's certification method has a checkable nonnegative tail. Set $S = 1 - k\nu_0$, $b = 1 - (k-1)\nu_0$, $r = S/b$, and $x_i = \nu_0 + Sy_i$, where $y_i \ge 0$ and $\sum_i y_i = 1$. Then

$$\frac{1}{x_1 \cdots x_k} = b^{-k} \sum_{j \ge 0} r^j h_j(1 - \mathbf{y}),$$

where h_j is the complete homogeneous polynomial. Since every $1 - y_i \in [0,1]$, truncating at degree N gives the integral-error bound

$$\frac{S^{k-1}}{(k-1)!\,b^k} \left[(1-r)^{-k} - \sum_{j=0}^{N} \binom{j+k-1}{k-1} r^j \right].$$

The expression is rational for rational ν_0. The truncated polynomial integrals over rational simplex caps can be evaluated exactly. A rigorous reproduction of the numerical enclosure must check both the finite integral values and these tails; floating-point agreement alone is insufficient. No such reproduction is claimed in this manuscript.

C Finite checks accompanying this note

The accompanying short Python program checks finite identities used in the new presentation: the partition identity and its sign-sensitive inequality, the unit-residue calculation of Lemma 5.2, the degree-twelve Stirling expansion, and the rational constants in the closing budget. The tests use integers and rational numbers. They are useful regression checks, not proofs of the asymptotic hypotheses, not an independent verification of the historical project programs, and not a numerical search for new twin primes. The mathematical proofs of the finite identities are given in the body of the note and do not depend on running the program. The supplied program passed 4,369 exact finite checks in both ordinary Python and Python with optimization enabled; the two JSON outputs were byte-identical.

D Provenance and responsible circulation

The main project sources used here are listed below with their dates and original filenames so that the author of a source record or a reviewer can locate the relevant proof. Their inclusion is not an endorsement of every historical status label. The latest degree-twelve report explicitly leaves the signed main term open. The older conditional dossier does not supply proofs of all its named analytic premises. No Lean build was run for this note.

This manuscript is an AI-assisted synthesis and mathematical audit of the supplied records. The known sieve construction belongs to Ford and Maynard; classical sieve and distribution results retain their original attribution. The note does not establish priority for a new theorem. It may be circulated as a *request for assessment of a conditional reduction*, not as a claimed proof of twin primes. It has not been submitted to any organization by its preparation.

References

[1] K. Ford and J. Maynard, *On the theory of prime-producing sieves*, arXiv:2407.14368v1 (2024), 107-page PDF. Numbering below follows that PDF. Definitions 4.3 and 4.5; equations (7.12)–(7.14); Lemmas 7.18, 7.20 and 7.21; Proposition 7.22; Theorem 8.3; equation (8.5) and the proof of Theorem 2.7(b), pp. 65–67. `https://arxiv.org/abs/2407.14368v1`.

[2] T. Tao, *254A, Notes 3: The large sieve and the Bombieri–Vinogradov theorem*, 10 January 2015. Theorem 17 and the maximal, divisor-weighted variants. `https://terrytao.wordpress.com/2015/01/10/254a-notes-3-the-large-sieve-and-the-bombieri-vinogradov-theorem/`.

[3] T. Tao, *254A, Notes 4: Some sieve theory*, 21 January 2015. Fundamental-lemma and positive upper-sieve tools. `https://terrytao.wordpress.com/2015/01/21/254a-notes-4-some-sieve-theory/`.

[4] D. R. Heath-Brown, *Lectures on sieves*, arXiv:math/0209360 (2002). `https://arxiv.org/abs/math/0209360`.

[5] Supplied project record, *HSTAR / post-R11 hostile audit and quantitative endgame repair*, 4 October 2026, `REPORT(20261004-110514).md`. Sections 3–7: pinned comparison, sparse boundary and the one-sided direct endgame; Section 8: the separate source-and-budget R11 export.

[6] Supplied project source audit, *Gate 1B: source-compatible margin repair and the corrected packet connector*, 12 October 2026, reproduced in `03_SOURCES_EVIDENCE.md`; and `REPORT(20261010-191142).md`, Section 3. Explicit shrunk g_ε versus unshrunk g_0 and prime-power split conventions.

[7] Supplied project record, *Gate 1B / Real R11: certified limiting Ford constant and the restored signed budget*, 24 October 2026, `REPORT(20261024-143157).md`. Sections 1–3: exact integral definition, reported rational enclosure, and the separate ε-boundary obligation.

[8] Supplied project record, *Gate 1B / Real R11: Global input-sieve stability of the original exterior*, 25 October 2026, `REPORT(20261025-131438).md`. Complete signed source transfer, with the original comparison unchanged.

[9] Supplied project record, *Gate 1B / Real R11: a stronger bound for the original A5 source*, 27 October 2026, `R11_Gate1B_Report_20261027.md`, Appendix A, "Complete proof of the cutoff bridge." The regional A5 bounds in the body are not used as a replacement for the complete exterior estimate.

[10] Supplied project record, *Real 1B / R11: post-first-success deduction*, 27 October 2026, `REPORT(20261027-054136).md`. Theorem 1 and Sections 5–6: proposed shrinking-roughness moments and degree-twelve source transfer; Section 9: explicit open signed main term.

[11] Supplied historical dossier, *From Full Physical R11 to the Twin Prime Conjecture*, 24 October 2026, `R11_Twin_Primes_Proof.pdf`. Conditional Gram/export and Lean-source discussion only. The direct scalar sign is rederived in Lemma 3.1 of the present note.